Saad Hussain

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Professional Summary

Computer Science Graduate and aspiring Data Scientist with strong experience in machine learning,AI,Data Science, natural language processing, and full-stack web development. Skilled in building intelligent systems including ML models, recommendation systems, and AI-powered chatbots. Experienced with tools such as Python, TensorFlow, PyTorch, Scikit-learn, LangChain,LangGraph and FastApi for developing real-world AI applications.

Core ICT Skills

Systems: Windows, Linux Basics, Software Installation
Networking: LAN/WAN, WiFi, Network Troubleshooting
Support: Incident Handling, Debugging, User Support
Hardware: Peripherals, Device Setup, Maintenance
Documentation: MS Word, MS Excel, Technical Reporting
Operations: APIs, Monitoring, Performance Optimization

Professional Experience

RegexByte IT Solutions | PSEB
AI Engineer Intern — *2025–May 2026*

- Developed agentic AI chatbot systems using LangChain and LangGraph, enabling multi-turn reasoning, context tracking, and tool-using LLM workflows.
- Designed and implemented a domain-specific Text-to-SQL system using Mistral-7B with PEFT/LoRA fine-tuning for natural language query understanding.
- Built an end-to-end AI assistant for a pet store use case that interacts conversationally with users, collects preferences, and dynamically transforms user intent into structured SQL queries.
- Integrated the LLM system with a product database, enabling real-time query execution and returning relevant product results within chat responses, including product details and direct URLs.
- Applied prompt engineering and state management techniques to maintain conversation memory and progressively refine user intent before query generation.
- Collaborated on Generative AI projects involving LLM integration, Retrieval-Augmented Generation (RAG), and evaluation of model outputs for accuracy and reliability.

National Center of Artificial Intelligence (NCAI), UET Peshawar — **Peshawar, Pakistan**
AI Intern — *Sep 2025 – Dec 2025*

- Gained hands-on experience in Artificial Intelligence, Generative AI, and Natural Language Processing (NLP) through practical training and project development.
- Developed AI-powered applications and projects involving machine learning, prompt engineering, and large language models (LLMs).
- Built Retrieval-Augmented Generation (RAG) systems for intelligent document retrieval and question-answering applications.
- Worked with AI automation tools, particularly n8n, to design and implement automated workflows integrating AI models and external services.
- Explored and applied concepts of AI agents, workflow orchestration, and business process automation using modern AI frameworks.
- Collaborated on real-world AI projects, contributing to data preparation, model experimentation, system integration, and deployment workflows.

Elevvo Pathways (Remote) — **Cairo, Egypt**
ML Intern — *Aug 2025 – Sep 2025*

- Performed data preprocessing, feature engineering, and model training using Python, Pandas, and Scikit-learn.
- Applied machine learning algorithms including Logistic Regression, Decision Trees, and K-Means for predictive analysis.
- Worked with real-world datasets to identify patterns and build predictive models.
- Visualized results and documented insights to support data-driven decision making.

Relevant Technical Projects

AI Chatbot with Tools and Persistent Memory

- Developed an intelligent chatbot using LangChain, LangGraph, and Streamlit supporting conversation threads and tool usage.
- Implemented persistent memory and dynamic tool integration to perform tasks such as document analysis and information retrieval.
- Designed the system architecture to simulate a ChatGPT-like conversational AI experience.

RAG System Implementing Google Research's TurboQuant Vector Compression
- Implemented Google Research's TurboQuant vector quantization algorithm (ICLR 2026) via TurboVec to build a compressed vector index, paired with local sentence-transformers embeddings and a Groq-hosted LLM for grounded, citation-backed answers.
- Achieved 8–16x vector compression with near-lossless retrieval by engineering a quantized indexing pipeline (chunking, embedding, compressed storage with stable document IDs) and validating it with a custom recall@k evaluation harness benchmarked against exact brute-force search.
- Designed a modular CLI and Streamlit web interface supporting document ingestion, persistent index management, interactive chat with source citations, and live compression/recall benchmarking.

Smart Waste Classification System using Deep Learning
- Developed a Convolutional Neural Network (CNN) for automated waste classification using computer vision techniques and deep learning.
- Performed image preprocessing, data augmentation, and model training on a real-world waste management dataset to improve classification performance and model generalization.
- Evaluated model performance using standard classification metrics and optimized the architecture for accurate multi-class waste categorization.
- Built an interactive web application using Gradio, enabling users to upload waste images and receive real-time classification results.
- Utilized Python, TensorFlow/Keras, OpenCV, Gradio, and Hugging Face to develop an end-to-end AI-powered waste management solution.

Education

Government Post Graduate College, Mardan
Bachelor's in Computer Science *2021–2025*

Certifications

5-Day AI Agents: Intensive Vibe Coding Course: Google,2026
Claude 101: Introduction to Building with AI: Anthropic, 2026
Get Started with Python: Google / Coursera, 2025
The Power of Statistics: Google / Coursera, 2025
Foundations of Data Science: Google / Coursera, 2025
Go Beyond Numbers: Translate Data Into Insights: Google / Coursera, 2025
The Nuts And Bolts of Machine Learning: Google / Coursera, 2025
Regression Analysis: Simplify Complex Data Relationship: Google / Coursera, 2025